

Horizontal Projectile Motion

Classwork, home assignments and the weekly assessment — fully worked, step by step.

Grade 11 · Egyptian Baccalaureate

28 questions

Step-by-step solutions

 $g = 10 \text{ m/s}^2$ unless stated**Mr. Gemy**
PHYSICS

The toolkit for this lesson

$$a_x = 0, a_y = g$$

the two motions are independent

$$h = \frac{1}{2} g t^2$$

vertical distance fallen

$$t = \sqrt{2h / g}$$

time of flight — height only

$$x = v t$$

horizontal range

$$v_y = g t$$

vertical component at any instant

$$v = \sqrt{v^2 + v_y^2}$$

speed at any instant

The one idea behind the whole lesson. A horizontal projectile falls exactly like a body dropped from rest, while at the same time it travels horizontally with a **constant** velocity. So the **time of flight depends on the height only**, never on the launch speed.

Horizontal projectile motion

CONCEPT

- 1 The figure shows the path of a projectile fired horizontally from a cannon with an initial velocity v until it reaches the point B.
- At which of the points 1, 2 and 3 is the **horizontal** component of the velocity the greatest? Explain.
 - At which of the points 1, 2 and 3 is the **vertical** component of the velocity the greatest? Explain.
 - Write the relation used to calculate the horizontal distance AB.
 - Write the relation used to calculate the vertical distance CB.
 - If the projectile is fired from the same height with a doubled speed ($2v$), what happens to: (1) the time it takes to reach the ground? (2) its horizontal range?

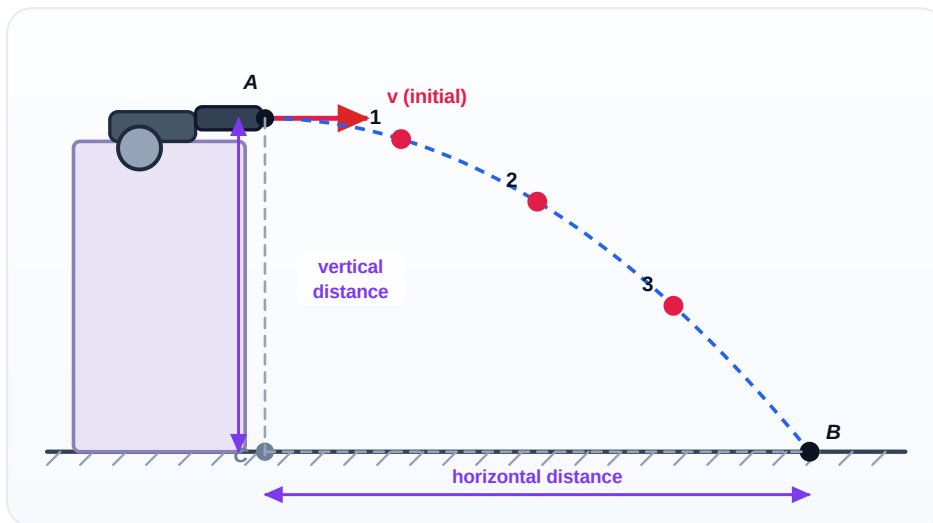


FIGURE 1 · THE PATH OF A HORIZONTAL PROJECTILE

PROBLEM

- 2 A body is projected horizontally with a velocity of 14.7 m/s from a height of 19.6 m above the ground. Taking $g = 9.8 \text{ m/s}^2$, find:
- the time the body takes to reach the ground, in seconds.
 - the horizontal range of the body, in metres.
 - the speed of the body just before it strikes the ground, in m/s .

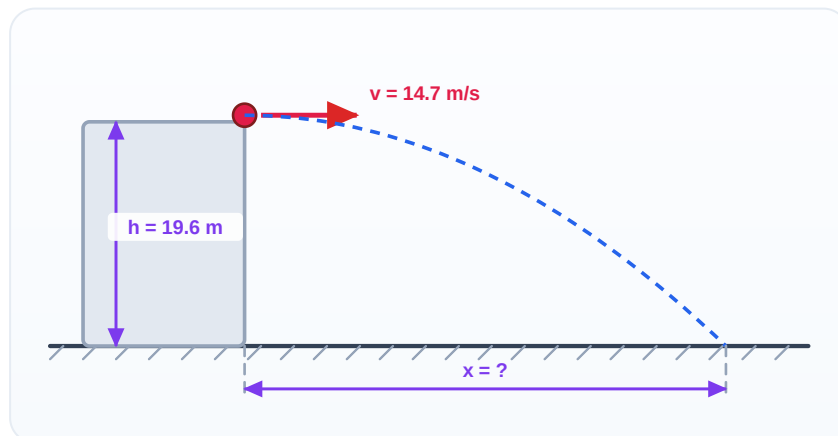


FIGURE 2

B The two independent motions

- 3 The **horizontal** motion of a projectile launched horizontally from a certain height is characterised by:

MCQ

- A a constant acceleration
- B a constant velocity
- C an increasing acceleration
- D an increasing velocity

- 4 Doubling the initial horizontal velocity of a horizontal projectile, while keeping the height constant, leads to:

MCQ

- A doubling the time of flight
- B halving the time of flight
- C doubling the horizontal range
- D increasing the horizontal range four times

- 5 A body is projected horizontally with a velocity of 4.0 m/s from a certain height and reaches the ground after 2.0 s . Calculate the height from which it was projected and its horizontal range.

PROBLEM

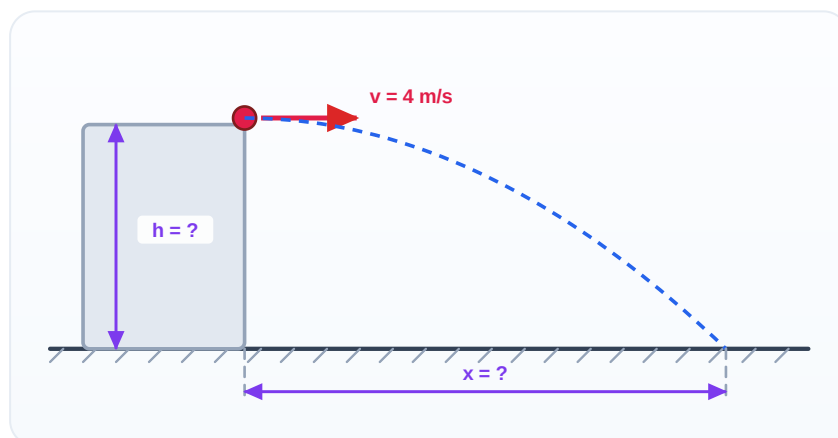


FIGURE 3

- 6 What happens to the horizontal range of a horizontally projected body if the **height** of the point of projection increases while the initial horizontal velocity stays constant? Explain.

EXPLAIN

- 7 A body is projected horizontally from the top of a building of height 45 m with an initial horizontal velocity of 20 m/s, neglecting air resistance and taking $g = 10 \text{ m/s}^2$. Calculate:

- 1) the time the body takes to reach the ground.
- 2) the horizontal distance covered before it reaches the ground.

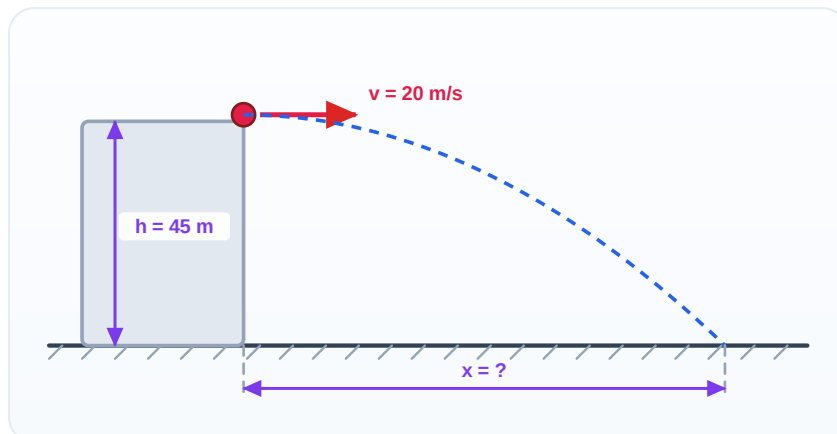


FIGURE 4

CLASSWORK 2

C The laws of horizontal projectiles

- 8 Two projectiles are fired horizontally from two different heights as follows:

RATIOS

- the first from a height of 20 m with a horizontal velocity of 10 m/s.
- the second from a height of 80 m with a horizontal velocity of 20 m/s.

Neglecting air resistance and taking $g = 10 \text{ m/s}^2$, **find the ratio between:**

- 1) the time of flight of the first projectile and that of the second.
- 2) the horizontal range of the first projectile and that of the second.

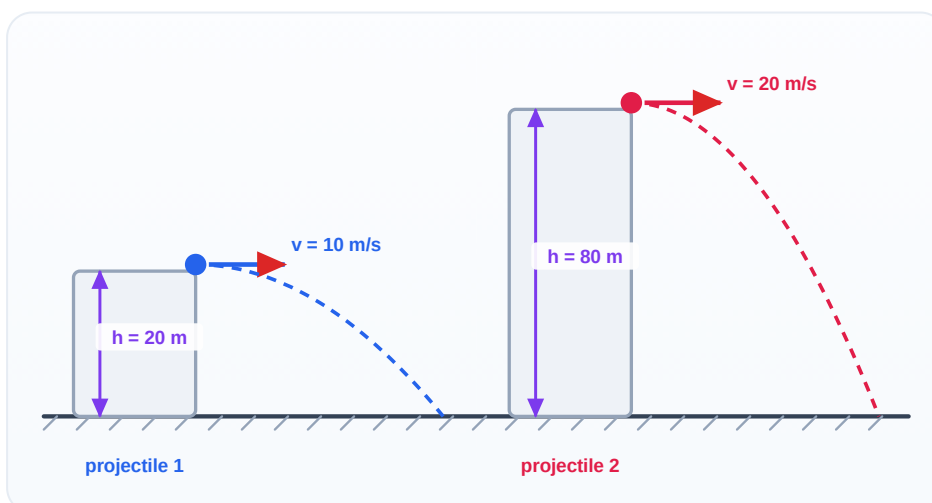


FIGURE 5

- 9 What happens to the time of flight and to the horizontal range of a body if it is projected horizontally from the **same height** but with a **greater** horizontal velocity? Explain.

EXPLAIN

More on the laws of horizontal projectiles

10 The motion of a body projected horizontally from a certain height is characterised by:

MCQ

- A zero acceleration during the horizontal motion
- B zero acceleration during the vertical motion
- C an acceleration equal to g during the horizontal motion
- D an acceleration equal to $\frac{1}{2}g$ during the vertical motion

11 What happens when the initial horizontal velocity of a horizontal projectile increases while the height is kept constant?

MCQ

- A the time of flight increases
- B the time of flight stays constant
- C the horizontal range stays constant
- D the horizontal range decreases

12 Two projectiles are fired horizontally from two different heights: the first from a height h with an initial horizontal velocity v , and the second from a height $4h$ with an initial horizontal velocity $2v$. Taking g as constant, find the ratio between:

RATIOS

- 1) the time of flight of the first projectile and that of the second.
- 2) the horizontal range of the first projectile and that of the second.

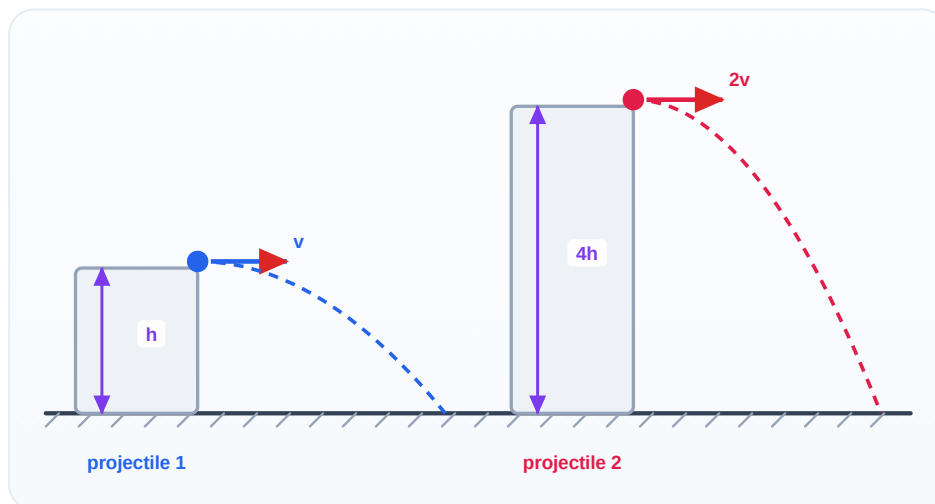


FIGURE 6

- 13 A billiard ball rolls off a table of height 0.60 m with an initial horizontal velocity of 2.4 m/s. Find the time the ball takes to hit the floor, and the horizontal distance between the edge of the table and the point where the ball lands.

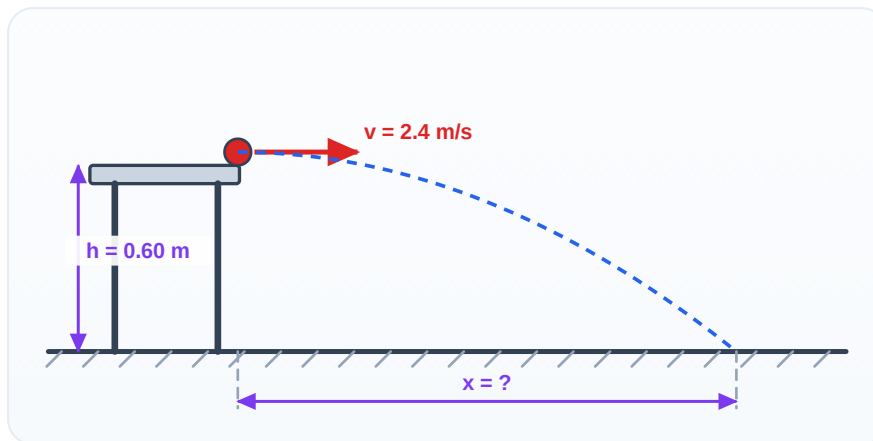


FIGURE 7

WEEKLY ASSESSMENT 2

E

Groups A, B and C

Group A

- 14 A body is projected horizontally from the **fourth** floor of a building of ten floors. If the same body is projected horizontally from the **tenth** floor with the same initial horizontal velocity, then the time to reach the ground and the horizontal range respectively:

MCQ

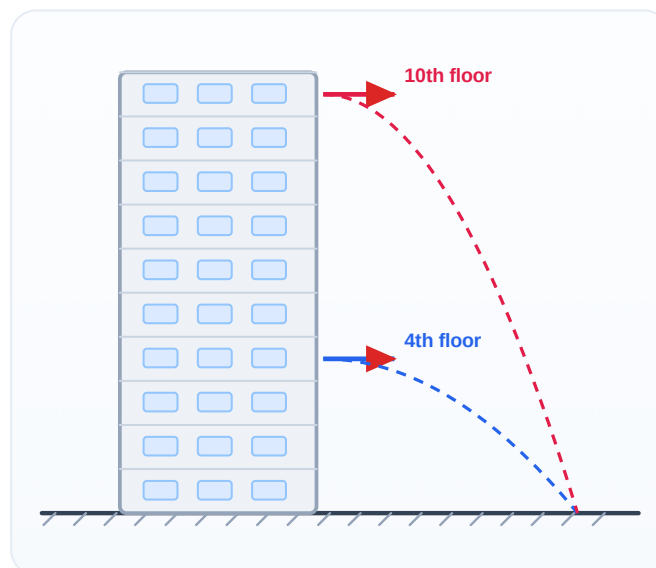


FIGURE 8

A stay constant · stay constant

B stay constant · increase

C increase · increase

D increase · stay constant

- 15 A helicopter flies horizontally with a velocity v at a height h and drops a food package so that it lands on a man standing on the ground. What is the distance between the helicopter and the man at the instant the package is released?

MCQ

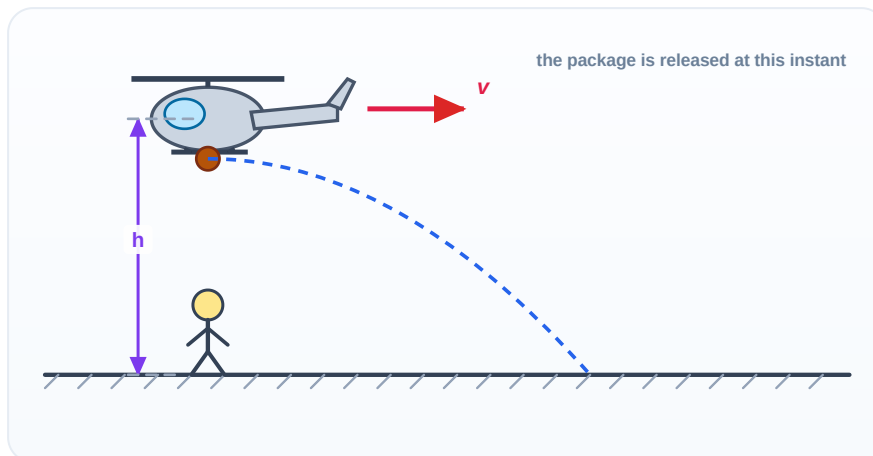


FIGURE 9

A $\sqrt{(2gh / v^2 + h^2)}$

B $\sqrt{(2v^2h / g + h^2)}$

C $\sqrt{(2ghv^2 + h^2)}$

D $\sqrt{(2ghv^2 + 1) / h^2}$

- 16 A body is projected horizontally with a velocity of 6.0 m/s from a certain height and reaches the ground after 3.0 s. Calculate the height from which it was projected and its horizontal range.

PROBLEM

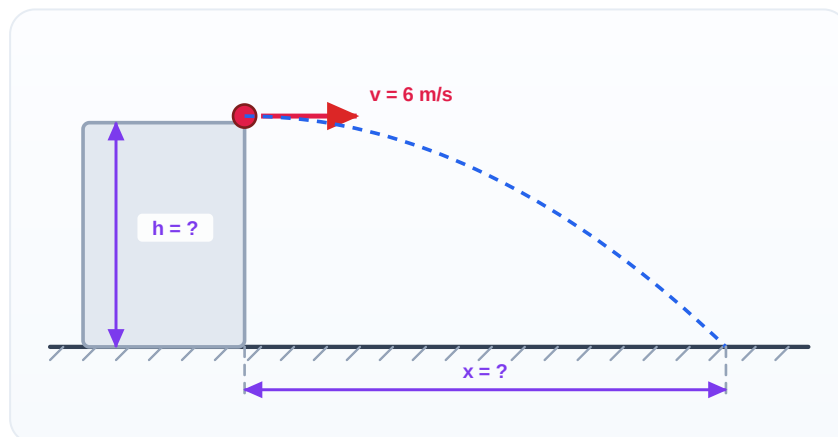


FIGURE 10

- 17 What happens to the time of flight and to the horizontal range of a body if it is projected horizontally from the same height but with a **smaller** horizontal velocity? Explain.

EXPLAIN

- 18** A billiard ball rolls off a table of height 0.75 m with an initial horizontal velocity of 1.5 m/s. What is the horizontal distance between the edge of the table and the point where the ball lands?

PROBLEM

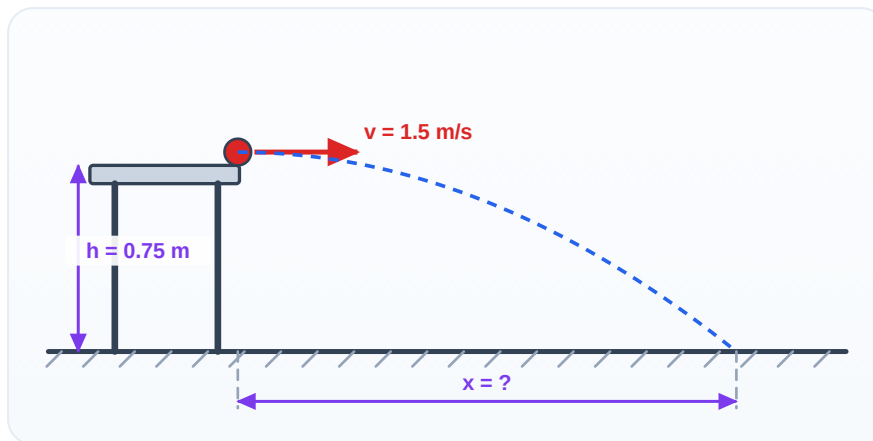


FIGURE 11

Group B

- 19** A body is projected horizontally from a certain height with an initial velocity v . If the initial horizontal velocity is doubled while the height is kept constant, the time the body takes to reach the ground:

MCQ

- A** is doubled
- B** is halved
- C** does not change
- D** becomes four times as large

- 20** The motion of a body projected horizontally from a certain height is characterised by:

MCQ

- A** a positive acceleration during the horizontal motion
- B** a negative acceleration during the horizontal motion
- C** an acceleration equal to g during the vertical motion
- D** an acceleration equal to $\frac{1}{4}g$ during the vertical motion

- 21** A body is projected horizontally from a height h with an initial horizontal velocity v , neglecting air resistance. What happens to the time it takes to reach the ground and to its horizontal range if the height is increased to $4h$ while the initial horizontal velocity stays constant? Explain.

EXPLAIN

- 22 The figure shows two coins A and B. Coin A is projected horizontally with an initial velocity v_0 , while coin B is released to fall freely from the same height at the same instant. **Which coin reaches the ground first?** Explain.

EXPLAIN

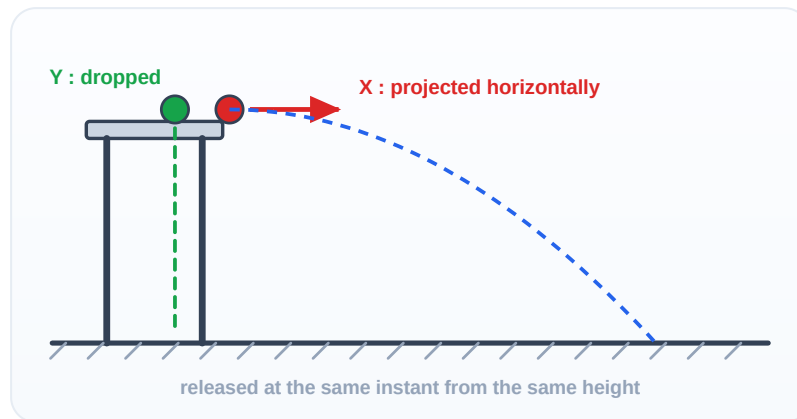


FIGURE 12

- 23 A body is projected horizontally with a velocity of 8 m/s from a height of 30 m above the ground. Taking $g = 10 \text{ m/s}^2$, find:

PROBLEM

- the time the body takes to reach the ground, in seconds.
- the horizontal range of the body, in metres.

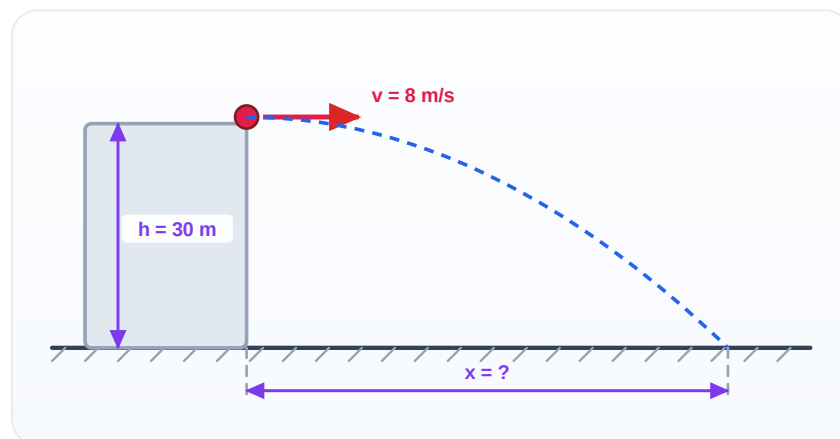


FIGURE 13

Group C

- 24 Halving the initial horizontal velocity of a horizontal projectile, while keeping the height constant, leads to:

MCQ

- | | |
|--|---|
| ☐ A halving the time of flight | ☐ B doubling the time of flight |
| ☐ C halving the horizontal range | ☐ D doubling the horizontal range |

25 The motion of a body projected horizontally from a certain height, while it is falling towards the ground, is characterised by:

MCQ

- A its horizontal velocity increases with time
- B its horizontal velocity decreases with time
- C its vertical velocity decreases with time
- D its vertical velocity increases with time

26 A ball is kicked horizontally from the top of a hill of height 22.0 m and lands at a horizontal distance of 35.0 m from the edge of the hill. **What is its initial horizontal velocity?**

PROBLEM

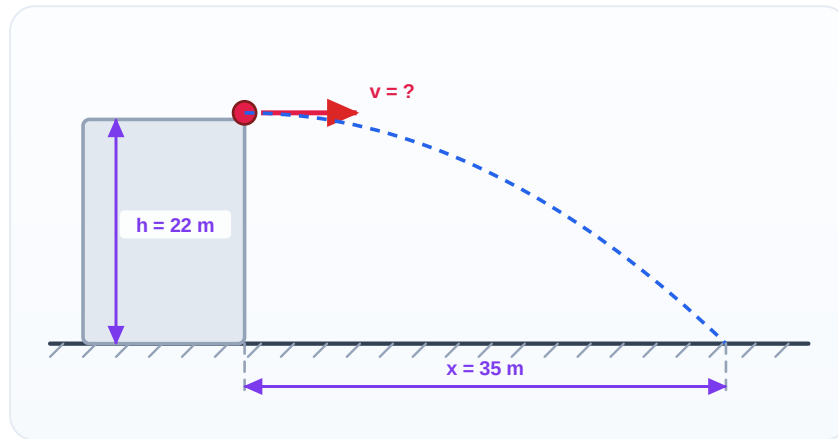


FIGURE 14

27 What happens to the time the body takes to reach the ground and to its horizontal range if it is projected horizontally from a **greater height** while its initial horizontal velocity stays constant? Explain.

EXPLAIN

28 A body is projected horizontally with a velocity of 3.0 m/s from a certain height and reaches the ground after 3.0 s. Find its horizontal range (x) and the height (h) from which it was projected.

PROBLEM

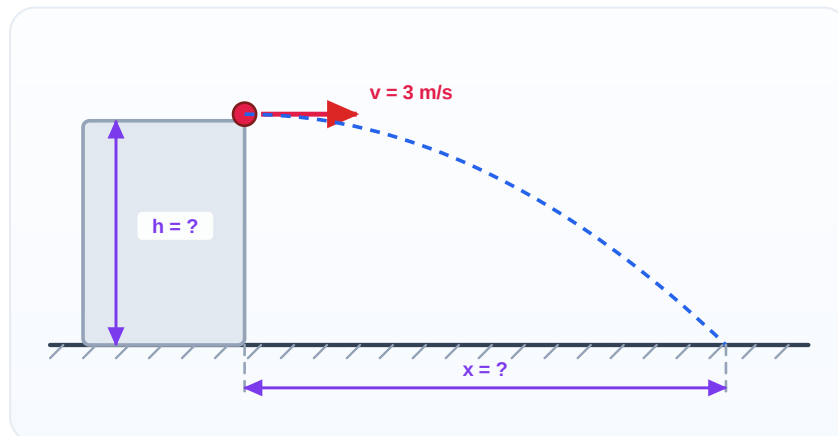


FIGURE 15



ANSWER KEY

Solutions — step by step

Every question is solved in full. Remember the golden rule: **the vertical motion decides the time, the horizontal motion decides the range.**

Part A — Classwork 1

1 reading the path of a horizontal projectile

- a) no force acts along the horizontal direction $\rightarrow a(x) = 0$
so $v(x)$ is the same at 1, 2 and 3 (and at every point of the path)
- b) the vertical motion is a free fall : $v(y) = g t$
 t is largest at point 3, so $v(y)$ is greatest at point 3
- c) $AB = v \times t$ (uniform horizontal motion)
d) $CB = \frac{1}{2} g t^2$ (free fall from rest, vertically)
- e) 1) $t = \sqrt{(2h / g)}$ does not contain $v \rightarrow$ the time does not change
2) $x = v t$, with t unchanged $\rightarrow$ the range is doubled

This is the heart of the lesson: the two motions are completely independent, and only the height decides how long the flight lasts.

a) equal at 1, 2, 3 · b) point 3 · e) time unchanged, range doubled

2 projectile from 19.6 m at 14.7 m/s ($g = 9.8 \text{ m/s}^2$)

- a) $h = \frac{1}{2} g t^2 \rightarrow t = \sqrt{(2h / g)} = \sqrt{(2 \times 19.6 / 9.8)} = \sqrt{4} = 2 \text{ s}$
- b) $x = v t = 14.7 \times 2 = 29.4 \text{ m}$
- c) $v(y) = g t = 9.8 \times 2 = 19.6 \text{ m/s}$
 $v = \sqrt{(v^2 + v(y)^2)} = \sqrt{(14.7^2 + 19.6^2)} = \sqrt{600.25} = 24.5 \text{ m/s}$

a) 2 s · b) 29.4 m · c) 24.5 m/s

Part B — Home assignment 1

3 the horizontal motion

$a(x) = 0 \rightarrow$ the horizontal velocity never changes

B) a constant velocity

4 doubling the launch speed

$t = \sqrt{(2h / g)}$ is independent of $v \rightarrow t$ stays the same
 $x = v t \rightarrow$ doubling v doubles x

C) doubling the horizontal range

5 height and range from the time of flight

$$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times (2)^2 = 20 \text{ m}$$

$$x = v t = 4 \times 2 = 8 \text{ m}$$

$$h = 20 \text{ m}, x = 8 \text{ m}$$

6 raising the point of projection

$$t = \sqrt{2h / g} \rightarrow \text{a larger } h \text{ gives a larger } t$$

$$x = v t, \text{ with } v \text{ unchanged} \rightarrow x \text{ increases}$$

The body keeps the same horizontal speed, but it now stays in the air for a longer time, so it travels further before landing.

the horizontal range increases

7 projectile from a 45 m building

$$1) t = \sqrt{2h / g} = \sqrt{2 \times 45 / 10} = \sqrt{9} = 3 \text{ s}$$

$$2) x = v t = 20 \times 3 = 60 \text{ m}$$

$$t = 3 \text{ s}, x = 60 \text{ m}$$

Part C — Classwork 2

8 two projectiles from 20 m and 80 m

$$\text{first : } t_1 = \sqrt{2 \times 20 / 10} = \sqrt{4} = 2 \text{ s} \quad x_1 = 10 \times 2 = 20 \text{ m}$$

$$\text{second : } t_2 = \sqrt{2 \times 80 / 10} = \sqrt{16} = 4 \text{ s} \quad x_2 = 20 \times 4 = 80 \text{ m}$$

$$1) t_1 : t_2 = 2 : 4 = 1 : 2$$

$$2) x_1 : x_2 = 20 : 80 = 1 : 4$$

$$1) 1 : 2 \cdot 2) 1 : 4$$

9 same height, greater speed

$$t = \sqrt{2h / g} \text{ does not contain } v \rightarrow \text{the time of flight does not change}$$

$$x = v t \rightarrow \text{the range increases in the same ratio as } v$$

Two bodies leaving the same table with different speeds hit the floor at the same instant — only the landing points are different.

the time stays the same, the range increases

Part D — Home assignment 2

10 the acceleration of a horizontal projectile

$$\text{horizontally : no force} \rightarrow a(x) = 0$$

$$\text{vertically : the weight only} \rightarrow a(y) = g$$

A) zero acceleration during the horizontal motion

11 increasing the launch speed

$t = \sqrt{(2h / g)}$ is independent of v

B) the time of flight stays constant

12 the two projectiles h, v and $4h, 2v$

$$1) \quad t_1 = \sqrt{(2h / g)} \quad t_2 = \sqrt{(2 \times 4h / g)} = 2 \sqrt{(2h / g)}$$

$$t_1 : t_2 = 1 : 2$$

$$2) \quad x_1 = v t_1 \quad x_2 = 2v \times t_2 = 2v \times 2 t_1 = 4 v t_1$$

$$x_1 : x_2 = 1 : 4$$

The time follows the **square root** of the height, so four times the height only doubles the time; the range then gains a second factor of 2 from the doubled speed.

1) $1 : 2 \cdot 2) 1 : 4$

13 the billiard ball

$$t = \sqrt{(2h / g)} = \sqrt{(2 \times 0.60 / 10)} = \sqrt{0.12} = 0.35 \text{ s}$$

$$x = v t = 2.4 \times 0.35 = 0.83 \text{ m}$$

$t = 0.35 \text{ s}, x = 0.83 \text{ m}$

Part E — Weekly assessment 2 · Group A

14 from the tenth floor instead of the fourth

the tenth floor is higher $\rightarrow h$ increases

$t = \sqrt{(2h / g)} \rightarrow t$ increases

$x = v t$, with v unchanged $\rightarrow x$ increases as well

C) both the time and the range increase

15 the helicopter and the man

the package needs $t = \sqrt{(2h / g)}$ to fall

in that time it travels horizontally $x = v t = v \sqrt{(2h / g)}$

at the instant of release the helicopter is exactly above the release point, so the distance to the man is the hypotenuse of a right angle :

$$d = \sqrt{(x^2 + h^2)} = \sqrt{(v^2 \times 2h / g + h^2)} = \sqrt{(2v^2h / g + h^2)}$$

B) $\sqrt{(2v^2h / g + h^2)}$

16 height and range from $t = 3 \text{ s}$

$$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 9 = 45 \text{ m}$$

$$x = v t = 6 \times 3 = 18 \text{ m}$$

$h = 45 \text{ m}, x = 18 \text{ m}$

17 same height, smaller speed

$$t = \sqrt{(2h / g)} \rightarrow \text{unchanged}$$

$$x = v t \rightarrow \text{smaller } v \text{ gives a smaller } x$$

the time stays the same, the range decreases

18 the billiard ball from 0.75 m

$$t = \sqrt{(2 \times 0.75 / 10)} = \sqrt{0.15} = 0.387 \text{ s}$$

$$x = v t = 1.5 \times 0.387 = 0.58 \text{ m}$$

x = 0.58 m (t = 0.39 s)

Group B

19 doubling the speed

$$t = \sqrt{(2h / g)} \text{ contains } h \text{ and } g \text{ only}$$

C) it does not change

20 the vertical acceleration

$$\text{the only force acting is the weight, directed downwards} \rightarrow a(y) = g = 10 \text{ m/s}^2$$

C) an acceleration equal to g during the vertical motion

21 raising the height from h to 4h

$$t_1 = \sqrt{(2h / g)} \quad t_2 = \sqrt{(2 \times 4h / g)} = 2 \sqrt{(2h / g)} = 2 t_1$$

$$x_1 = v t_1 \quad x_2 = v t_2 = 2 v t_1 = 2 x_1$$

Four times the height does not give four times the time, because the time depends on the **square root** of the height.

both the time and the range are doubled

22 the projected coin and the dropped coin

vertically the two coins start with $v(y) = 0$ and have the same $a = g$ and the same h

$$h = \frac{1}{2} g t^2 \text{ gives the same } t \text{ for both}$$

the horizontal velocity of A does not affect its vertical motion

This is the classic experiment that proves the independence of the two motions: A lands far away from the table, B lands just below it, but both land at the same moment.

the two coins reach the ground at the same instant

23 projectile from 30 m at 8 m/s

$$\text{a) } t = \sqrt{(2h / g)} = \sqrt{(2 \times 30 / 10)} = \sqrt{6} = 2.45 \text{ s}$$

$$\text{b) } x = v t = 8 \times 2.45 = 19.6 \text{ m}$$

a) 2.45 s · b) 19.6 m

Group C

24 halving the launch speed

t is independent of $v \rightarrow$ unchanged

$x = v t \rightarrow$ halving v halves x

C) halving the horizontal range

25 the velocity during the fall

$v(x) = v = \text{constant}$ (no horizontal force)

$v(y) = g t \rightarrow$ grows steadily with time

D) its vertical velocity increases with time

26 finding the launch speed from the range

$t = \sqrt{(2h / g)} = \sqrt{(2 \times 22 / 10)} = \sqrt{4.4} = 2.10 \text{ s}$

$x = v t \rightarrow v = x / t = 35 / 2.10 = 16.7 \text{ m/s}$

$v = 16.7 \text{ m/s}$

27 projecting from a greater height

$t = \sqrt{(2h / g)} \rightarrow$ a larger h gives a larger t

$x = v t$, with v unchanged $\rightarrow x$ increases too

both the time of flight and the horizontal range increase

28 range and height from $t = 3 \text{ s}$

$x = v t = 3 \times 3 = 9 \text{ m}$

$h = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 9 = 45 \text{ m}$

$x = 9 \text{ m}$, $h = 45 \text{ m}$